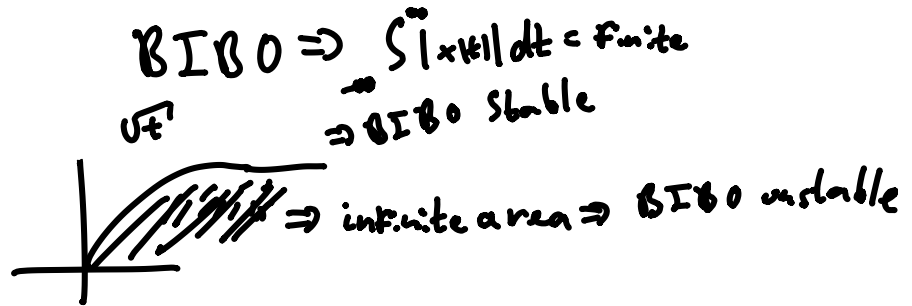


Worksheet 6



1) An example BIBO stable system impulse response is

a. $e^t u(t) \Rightarrow \text{exponential unbounded output} \Rightarrow BIBO \text{ unstable}$

b. $\sqrt{t} u(t) \Rightarrow BIBO \text{ unstable} \Rightarrow \text{infinite area}$

c. $e^{-t} u(t) \Rightarrow \text{exponential decaying} \Rightarrow \text{finite area} \Rightarrow BIBO \text{ stable}$

d. $e^t [u(t) - u(t-1)]$

$\Rightarrow BIBO \text{ stable}$

2) Which system below is causal?

a. $y(t) = \frac{1}{2}x(t+1) + \frac{1}{2}x(t-1)$

For $x(t) < 0, x(t) = 0$

b. $y(t+1) = \frac{1}{2}x(t+1) + \frac{1}{2}x(t)$

For $x(t) = 0, y(t) = 0$

c. $y(t-1) = \frac{1}{2}x(t) + \frac{1}{2}x(t-2)$

b. Depends on present/past input $\Rightarrow \text{causal}$

d. $y(t) = 2x(t+1)$

3) Which system below is causal?

a. $\frac{d^2 y}{dt^2} - 3 \frac{dy}{dt} + y(t-1) = \frac{d^2 x(t-1)}{dt^2} + 2 \frac{dx}{dt}$

$\Rightarrow \text{Depends on present/past input} \Rightarrow \text{causal}$

b. $\frac{d^2 y}{dt^2} - 3 \frac{dy}{dt} + y(t) = \frac{d^2 x(t)}{dt^2} + 2 \frac{dx(t+1)}{dt}$

c. $\frac{d^2 y(t-1)}{dt^2} - 3 \frac{dy}{dt} + y(t+1) = \frac{d^2 x(t)}{dt^2} + 2 \frac{dx}{dt}$

d. $\frac{d^2 y}{dt^2} - 3 \frac{dy}{dt} + y(t-1) = \frac{d^2 x(t)}{dt^2} + 2 \frac{dx(t+1)}{dt}$

4) A system with impulse response $h(t)$ is defined as BIBO stable if _____.

a. $|\int_{-\infty}^{\infty} h(t) dt| < \infty$

b. $\int_{-\infty}^{\infty} h(t) dt < \infty$

c. $\int_{-\infty}^{\infty} |h(t)| dt < \infty$

d. $|h(t)| \leq 1$ for all t

5) A system with impulse response $h(t) = 3e^{-5t}u(t)$ has frequency response $H(\omega) =$ _____.

a. $\frac{5}{3+j\omega}$

b. $\frac{15}{j\omega}$

c. $\frac{3}{5-j\omega}$

d. $\frac{3}{5+j\omega}$

$$\begin{aligned} \hat{H}(\omega) &= \int_{-\infty}^{\infty} 3e^{-5t}u(t)e^{j\omega t} dt \\ &= 3 \int_0^{\infty} e^{-(5-j\omega)t} dt = 3 \left[\frac{e^{-(5-j\omega)t}}{-(5-j\omega)} \right]_0^{\infty} \\ &= 3 \left(0 - \left(-\frac{1}{5-j\omega} \right) \right) = \frac{3}{5-j\omega} \end{aligned}$$

6) A system with impulse response $h(t) = e^{-5|t|}$ has frequency response $H(\omega) =$ _____.

a. $-\frac{10}{(5+j\omega)^2}$

b. $\frac{10}{25+\omega^2}$

c. $-\frac{10}{25-\omega^2}$

d. $-\frac{10}{25+\omega^2}$

$$\begin{aligned} \hat{H}(\omega) &= \int_{-\infty}^{\infty} e^{-5|t|} e^{j\omega t} dt \\ &= \int_{-\infty}^0 e^{-(5-j\omega)t} dt + \int_0^{\infty} e^{-(5+j\omega)t} dt \\ &= \left[\frac{e^{-(5-j\omega)t}}{-(5-j\omega)} \right]_{-\infty}^0 + \left[\frac{e^{-(5+j\omega)t}}{-(5+j\omega)} \right]_0^{\infty} \\ &= \left(0 - \left(-\frac{1}{5-j\omega} \right) \right) + \left(\frac{1}{5+j\omega} - 0 \right) \\ &= \frac{1}{5-j\omega} + \frac{1}{5+j\omega} = \frac{5-j\omega+5+j\omega}{25+\omega^2} = \frac{10}{25+\omega^2} \end{aligned}$$

7) A system defined by the LCCDE $\frac{d^3 y}{dt^3} - 2y(t) = \frac{d^2 x}{dt^2} - x(t)$ has frequency response $H(\omega) =$ _____.

a. $\frac{-1-\omega^2}{2+j\omega^3}$

b. $\frac{1+\omega^2}{2+j\omega^3}$

c. $\frac{1+\omega^2}{2+\omega^2}$

d. $\frac{1+j\omega}{2+j\omega^3}$

$$\begin{aligned} x(t) &= e^{j\omega t} \\ y(t) &= \hat{H}(\omega) e^{j\omega t} \\ \frac{dy}{dt} &= \hat{H}(\omega) j\omega e^{j\omega t} \\ \frac{d^2 y}{dt^2} &= \hat{H}(\omega) (j\omega)^2 e^{j\omega t} \\ \frac{d^3 y}{dt^3} &= \hat{H}(\omega) (j\omega)^3 e^{j\omega t} \\ \hat{H}(\omega) (j\omega)^3 e^{j\omega t} - 2\hat{H}(\omega) e^{j\omega t} &= (j\omega)^2 e^{j\omega t} - e^{j\omega t} \\ \hat{H}(\omega) (j\omega)^3 - 2\hat{H}(\omega) &= (j\omega)^2 - 1 \\ \hat{H}(\omega) (j\omega^3 - 2) &= -\omega^2 - 1 \\ \hat{H}(\omega) &= \frac{-\omega^2 - 1}{j\omega^3 - 2} = \frac{1 + \omega^2}{2 + j\omega^3} \end{aligned}$$

8) A system with a real impulse response function $h(t)$.
has frequency response at $\omega = 2$ rad/s $H(2) = 4/45^\circ$. $H(\omega)$ evaluated at $\omega = -2$ rad/s
is $H(-2) = \underline{\hspace{2cm}}$.

- a. $4/45^\circ$
- b. $4/135^\circ$
- c. $4/-45^\circ$
- d. $4/-135^\circ$

$$H(-\omega) = H^*(\omega)$$

$$\begin{aligned} x(t) &= e^{j\omega t} \\ y(t) &= \hat{H}(\omega) e^{j\omega t} \\ \frac{dy}{dt} &= \hat{H}(\omega) j\omega e^{j\omega t} \\ \frac{d^2 y}{dt^2} &= \hat{H}(\omega) (j\omega)^2 e^{j\omega t} \end{aligned}$$

9) The system with LCCDE $\frac{d^2 y}{dt^2} + 2y(t) = 3 \frac{dx}{dt} - 3x(t)$ and input $x(t) = \sqrt{2} \cos(t - 45^\circ)$
has output $y(t) = \underline{\hspace{2cm}}$.

- a. $6\sin(t)$
- b. $-6\sin(t)$
- c. $-6\cos(t + 135^\circ)$
- d. $6\cos(t + 135^\circ)$

$$\begin{aligned} &= \hat{H}(\omega) (j\omega)^2 e^{j\omega t} + 2\hat{H}(\omega) e^{j\omega t} = 3j\omega e^{j\omega t} - 3e^{j\omega t} \\ &= \hat{H}(\omega) (j\omega^2 + 2) = 3j\omega - 3 \\ &= \hat{H}(\omega) (-\omega^2 + 2) = 3j\omega - 3 \\ &\hat{H}(\omega) = \frac{3j\omega - 3}{-\omega^2 + 2} \end{aligned}$$

$$\begin{array}{r} 18 \\ / 1 \\ 2 \quad 9 \\ / 1 \\ 3 \quad 3 \end{array}$$

$$\begin{aligned} \hat{H}(1) &= \frac{3j - 3}{-1 + 2} = 3j - 3 \Rightarrow |\hat{H}(1)| = \sqrt{18} = 3\sqrt{2} \\ \tan^{-1}\left(\frac{3}{-3}\right) &= \tan^{-1}\left(-1\right) = -45^\circ + 180^\circ = 135^\circ \\ y(t) &= 3\sqrt{2} \cdot \sqrt{2} \cos(t - 45^\circ + 135^\circ) \\ y(t) &= -6\sin(t) \end{aligned}$$

10) Which LCCDE below represents a critically-damped system?

- a. $\frac{d^2 y}{dt^2} + 6 \frac{dy}{dt} + 4.5y(t) = \frac{dx}{dt} + 2x(t)$
- b. $\frac{d^2 y}{dt^2} + 2.25 \frac{dy}{dt} + 3y(t) = \frac{dx}{dt} + 2x(t)$
- c. $\frac{d^2 y}{dt^2} - 3 \frac{dy}{dt} + 2.25y(t) = \frac{dx}{dt} + 2x(t)$
- d. $\frac{d^2 y}{dt^2} + 3 \frac{dy}{dt} + 2.25y(t) = \frac{dx}{dt} + 2x(t)$

$$\begin{aligned} &2.25 \quad 2.25 \\ &\underline{2.25} \\ &0 \end{aligned}$$

$$|xj + y| \Rightarrow \sqrt{x^2 + y^2}$$

$$s^2 + 6s + 4.5 = 0$$

$$36 - 4 \cdot 1 \cdot 4.5 = 36 - 18 = 18 = + \text{ overdamped}$$

$$2.25^2 - 4 \cdot 3 = -6.94$$

stable b/c $\alpha_1, \alpha_2 > 0$

11) Which LCCDE below represents an unstable system?

- a. $\frac{d^2 y}{dt^2} + 3 \frac{dy}{dt} + y(t) = \frac{dx}{dt} - 2x(t)$
- b. $\frac{d^2 y}{dt^2} + 3 \frac{dy}{dt} + y(t) = \frac{dx}{dt} + 2x(t)$

c. $\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + y(t) = \frac{dx}{dt} + 2x(t)$

d. $-\frac{d^2y}{dt^2} - 3\frac{dy}{dt} - y(t) = -\frac{dx}{dt} - 2x(t)$

↑ stable b/c can div. all neg. out $\frac{3}{2} = 1$ b_1 b_2

12) The impulse response corresponding to LCCDE $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y(t) = \frac{dx}{dt} + 2x(t)$ is

$h(t) = \underline{\hspace{2cm}}$.

a. $2e^{-t}u(t)$

b. $(1+t)e^{-t}$

c. $(1+t)e^{-t}u(t)$

d. $(e^{-t} + e^t)u(t)$

Char. eqn $s^2 + 2s + 1 = 0$

$\sigma = 1$

$\omega_0 = 1$

$\rho_1 = -1$

$\rho_2 = -1$

$\sigma = \omega_0 \Rightarrow$ Crit. Damped

$h(t) = 1 \cdot \frac{dh(t)}{dt} + 2h(t)$

$h_c(t) = te^{-t}u(t)$
 $b_1 = 1$ $b_2 = 2$

$f'g + g'f = (fg)'$

$\frac{dh_c(t)}{dt} = 1e^{-t}u(t) + t(-e^{-t})u(t) + te^{-t}\delta(t)$

$C_2 = b_2 - \sigma b_1$

$C_1 = b_1 = 1$

$C_2 = 2 - 1 \cdot 1 = 1$

$h(t) = (C_1 + C_2 t)e^{-\sigma t}u(t)$
 $= (1 + t)e^{-t}u(t)$

$\alpha = \frac{a_1}{2}$

$\omega_0 = \sqrt{a_2}$

$\zeta = \frac{\alpha}{\omega_0} = \frac{a_1}{2\sqrt{a_2}}$

attenuation coefficient (Np/s),

undamped natural frequency (rad/s),

damping coefficient (unitless).

$p_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = \omega_0 [-\zeta + \sqrt{\zeta^2 - 1}]$

$p_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = \omega_0 [-\zeta - \sqrt{\zeta^2 - 1}]$